The galaxy-bias ladder under one-loop bias

The correlated-prior note redone with model A at $k_{\rm max} = 0.075\,h/{\rm Mpc}$

November 2026

1 Summary

The question. The forecast measures DESI galaxy clustering alone: the angular band powers C_ℓ^{gg} of thin redshift bins, computed without the Limber approximation, for six tracer groups (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four bin layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. The correlated-prior note (`notes/correlated_bias_prior`, "v1" below) reports it with linear galaxy bias, as a ladder of four stated bias assumptions tested against random bias histories. The companion note `notes/linear_bias_kmax` shows that linear bias is already biased at v1's cut, $k_{\rm max} = 0.075\,h/{\rm Mpc}$: fitted to a one-loop truth on the $\Delta d_{\rm com} = 62$ Mpc layout, it moves Ω_m by 2.0–2.4σ and (w_0, w_a) by 0.7–1.1σ under every linear-bias prior. This note redoes v1's two central results, the figure-of-merit ladder and the accuracy test, with the one-loop model A of `notes/higher_order_bias` on v1's kept band powers, so every difference from v1 is the bias model.

Symbols: $w(a) = w_0 + w_a(1-a)$ is the dark-energy equation of state, with a the scale factor. The figure of merit ${\rm FoM}(w_0, w_a) = [\det {\rm Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) error ellipse; larger is better. Ω_m is the matter density today in units of the critical density, A_s the amplitude of the primordial curvature power spectrum (sampled as $10^9 A_s$), and σ a parameter's marginalized one-sigma error. $D(z)$ is the linear growth factor, normalized to $D(0) = 1$.

The scale cut. $k_{\rm max}$ labels one cut per bin. Bin i is cut at $k_{\rm max}/D(\bar z_i)$, with $\bar z_i$ its mean redshift: matter is less nonlinear at higher redshift, so the cut moves to higher wavenumber k as $1/D$. The cut is the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, computed from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it. A band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; \alpha = 0.05, \tag{1}$$

and the kept bands are intersected with the shipped mask. The cut was built on the linear model. Every $k_{\rm max}$ in this note is the $z = 0$ value.

The linear-bias ladder. v1 writes the linear bias of bin i in group g as

$$b_{1,i} \;=\; \frac{B_g}{D(z_i)}\,(1+\delta_i), \qquad \delta_i \;=\; s_g\,(z_i - \bar z_g) + \varepsilon_i, \tag{2}$$

with z_i the bin redshift, B_g a free amplitude and s_g a free tilt per group (flat priors), $\bar{z}_g$ the group's mean bin redshift, and ε_i a residual. The residuals carry a Gaussian prior correlated within a group, $\langle \varepsilon_i \varepsilon_j \rangle = \sigma_g^2 \exp(-|z_i - z_j|/L)$, and zero across groups, with σ_g the per-bin width and L the correlation length in redshift. The ladder has four linear-bias priors. **Growth-tied**: one amplitude per group, with the bias following $1/D(z)$, $b_{1,i} = B_g/D(z_i)$, and no tilt or residual. The **baseline prior**: $\sigma_g = 3\%$ for BGS and 1% for the other groups, $L = 0.2$. The **pessimistic prior**: $\sigma_g = 3\%$ in every group, $L = 0.05$. **Free**: one $b_{1,i}$ per bin, each with a flat prior. v1's Summary gives the physical source of each number.

Model A. Model A keeps this per-bin linear bias and adds the one-loop terms. As computed here, the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_0\, k^2 P_{\rm mm}(k) \;+\; \frac{\rm SN_0}{\bar{n}_i}, \tag{3}$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ holds the one-loop terms. These carry the quadratic bias b_2, the response of the galaxy density to the square of the matter density contrast, and the tidal bias b_{s^2}, its response to the squared tidal field; the third-order bias b_3 is zero. α_0 is the k^2 counterterm, in $h^{-2}\,{\rm Mpc}^2$, which absorbs the sensitivity of the loops to small scales. $\rm SN_0$ is the white-noise power beyond Poisson, in units of the bin's Poisson power $1/\bar{n}_i$, with $\bar{n}_i$ the bin's number density. The higher-order biases follow each bin's own linear bias through calibrations, plus one offset per group:

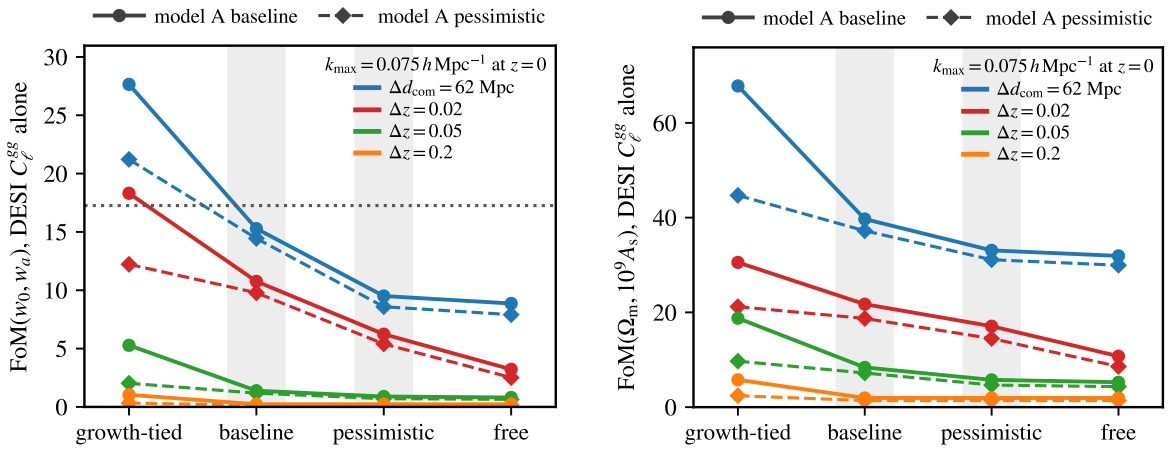
$$b_{2,i} \;=\; b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} \;=\; -\tfrac{4}{7}\,(b_{1,i} - 1) + \Delta_{s,g}, \tag{4}$$

with $b_2^{\rm L}$ the separate-universe fit of Lazeyras et al. [3] and the second relation the co-evolution of a population formed with local bias [4]. Each group carries one α_0 and one $\rm SN_0$. The priors are Gaussian and centered on zero, with widths 1 on Δ_2, 0.5 on Δ_s, $0.5/k_{\rm max}^2 = 88.9\,h^{-2}\,{\rm Mpc}^2$ on α_0 (a 50% correction to the linear spectrum at $k_{\rm max}$, the rule of the DESI DR1 full-shape analysis [5, 6] moved to this cut), and 2 on $\rm SN_0$. Redshift-space distortions are linear (Kaiser), with each bin's redshift distribution convolved by the sampled velocity dispersion σ_v (the Fingers-of-God smearing), as in v1.

Two variants of the higher-order sector sit on every linear-bias prior. **Model A baseline** has one Δ_2, Δ_s, α_0 and $\rm SN_0$ per group with the priors above, 24 parameters. **Model A pessimistic** also tilts the Δ_2, Δ_s and α_0 of each group linearly in redshift, with flat priors on the tilts, up to 42 parameters; $\rm SN_0$ never tilts, because its Poisson part is known bin by bin.

The figure of merit. Figure 1(a) gives the FoM of every linear-bias prior under both variants. The FoM falls from growth-tied to free under each variant; on $\Delta z = 0.2$ every group holds one or two bins, the free amplitude and tilt already free every bin, and the baseline, pessimistic and free priors coincide. Model A baseline on the baseline prior gives a FoM of about 15, 11, 1.4 and 0.2 (layouts in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2), and model A pessimistic on the pessimistic prior about 9, 5, 0.7 and 0.15. On $\Delta d_{\rm com} = 62$ Mpc the first, about 15, sits below the LSST Y1 3x2pt reference of 17.26 [7]; under model A only the growth-tied prior, which the accuracy test below excludes, exceeds that reference.

Figure 1(b) gives the figure of merit of the matter density and the primordial amplitude, $\mathrm{FoM}(\Omega_m, 10^9 A_s) = [\det \mathrm{Cov}(\Omega_m, 10^9 A_s)]^{-1/2}$ in the sampled units, from the same Fisher inverse. Under model A baseline it falls from growth-tied to free from about 68 to 32 on



(a) (b)

Figure 1: **(a)** FoM(w_0, w_a) and **(b)** FoM($\Omega_m, 10^9 A_s$) of DESI C_ℓ^{gg} alone under the four linear-bias priors. Colors: the bin layouts. Solid lines: model A baseline; dashed lines: model A pessimistic. Gray columns: the baseline and pessimistic priors. Dotted line in (a): the LSST Y1 3x2pt FoM(w_0, w_a), 17.26. $k_{max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

$\Delta d_{com} = 62$ Mpc, from 31 to 11 on $\Delta z = 0.02$ and from 19 to 5 on $\Delta z = 0.05$; on $\Delta z = 0.2$ it falls from about 6 to 2 and then stays flat. Model A pessimistic on the pessimistic prior gives about 31, 14, 5 and 1.4. Appendix E sets these numbers next to v1's linear bias and to exact calibrations, neither of which is a result here.

The accuracy test. The truths are v1's: every bin's linear bias follows its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere) plus a residual drawn from the baseline or the pessimistic prior, 2000 truths per case with v1's random seed. Under model A the truth's b_2 and b_{s^2} follow its b_1 through Eq. (4). Every linear-bias prior fits every truth under both variants, centered on the growth-tied shape. Figure 2 draws the (w_0, w_a) shift of each fit in units of its own ellipse. Where a fit's prior equals the truth's scatter (the baseline prior in the top panel, the pessimistic prior in the middle one), the mean shift is not a bias: that fit's ellipse already covers this scatter. Its bottom panel shows the calibration drift discussed below.

The verdict is a goodness of fit of the parameter recovery,

$$\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}, \tag{5}$$

where χ^2 is the squared distance of the recovered parameters from the truth in units of the fit's own covariance, with $n_{\mathrm{dof}} = 2$ for (w_0, w_a) and $n_{\mathrm{dof}} = 1$ for Ω_m or A_s, and $\langle \cdot \rangle$ averages over the random truths and the data noise. It is 1 when the error bars cover the truth as often as they claim; above 1 it is the factor by which they are too small, and below 1 they are conservative. It scores the recovery of the parameters, not the χ^2 of the data: a biased fit can match the band powers well while its cosmology is wrong. v1's appendix "The accuracy test:

the figure of bias" derives both statistics, and Figure 3 draws the goodness of fit of model A baseline under the pessimistic truths.

The v1 verdict repeats under model A (Figure 3).

- *Growth-tied is excluded.* Its ellipse is 5.6–10.3 times too small under model A baseline and 2.0–4.4 times under model A pessimistic.
- *The baseline prior is honest under its own scatter*, with a goodness of fit of 1.00–1.01, but 1.55–2.3 times too small under pessimistic scatter (model A baseline; 1.6–2.35 under model A pessimistic). The exception is $\Delta z = 0.2$, at 1.00, because its groups hold one or two bins and the free amplitude and tilt already free every bin.
- *The pessimistic prior is honest* under pessimistic scatter (at most 1.02) and conservative under baseline scatter (0.85–1.00).
- Ω_m *and* A_s *agree.* Under model A baseline the growth-tied fit's Ω_m error bar is 3.9–10.2 times too small; under pessimistic scatter the baseline-prior fit's is 1.29–2.66 times too small (A_s: 1.15–1.81; both 1 on $\Delta z = 0.2$), and the pessimistic-prior fit's at most 1.04 (A_s: 1.01).

A drift of the calibrations. The random truths move only the linear bias; the higher-order biases follow it through calibrations assumed exact up to one constant offset per group. A third truth tests that assumption. It drifts the offsets linearly with the bin redshift across each group, Δ_2 from about -0.5 to $+0.5$ and Δ_s from about -0.25 to $+0.25$ (half the prior width at each end), and keeps b_1 at its fiducial. Under model A baseline this drift shifts (w_0, w_a) by 1.0–1.8σ under the growth-tied prior. Under the baseline, pessimistic and free priors it gives 0.35–0.57σ on $\Delta d_{\rm com} = 62$ Mpc and 0.50–0.53σ on $\Delta z = 0.05$ and 0.2, all above $\sigma/3$, and 0.07–0.15σ on $\Delta z = 0.02$.

Model A pessimistic absorbs this drift exactly, by construction: its tilts have the drift's shape. The test shows only that the tilts work for a linear drift. A nonlinear drift of the calibrations within a group is untested. Protecting against the linear drift costs little on the thin layouts: model A pessimistic on the baseline prior gives about 14 and 10 against 15 and 11 for model A baseline.

Recommendation. Quote the forecast, as v1 does for linear bias, as a ladder of two stated configurations.

- **Reasonable**: model A baseline on the baseline prior, a FoM of about 15, 11, 1.4 and 0.2 on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2. It assumes two things: each bin's linear-bias residual is no larger and no rougher than the baseline prior, and the calibrations of b_2 and b_{s^2} hold within a group up to one constant offset. If the first fails at the pessimistic size, the ellipse is 1.55–2.3 times too small; if the second fails by the linear drift above, (w_0, w_a) moves by about 0.5σ on three layouts.
- **Pessimistic**: model A pessimistic on the pessimistic prior, about 9, 5, 0.7 and 0.15. It is honest under both families of random truths and absorbs a linear calibration drift exactly.

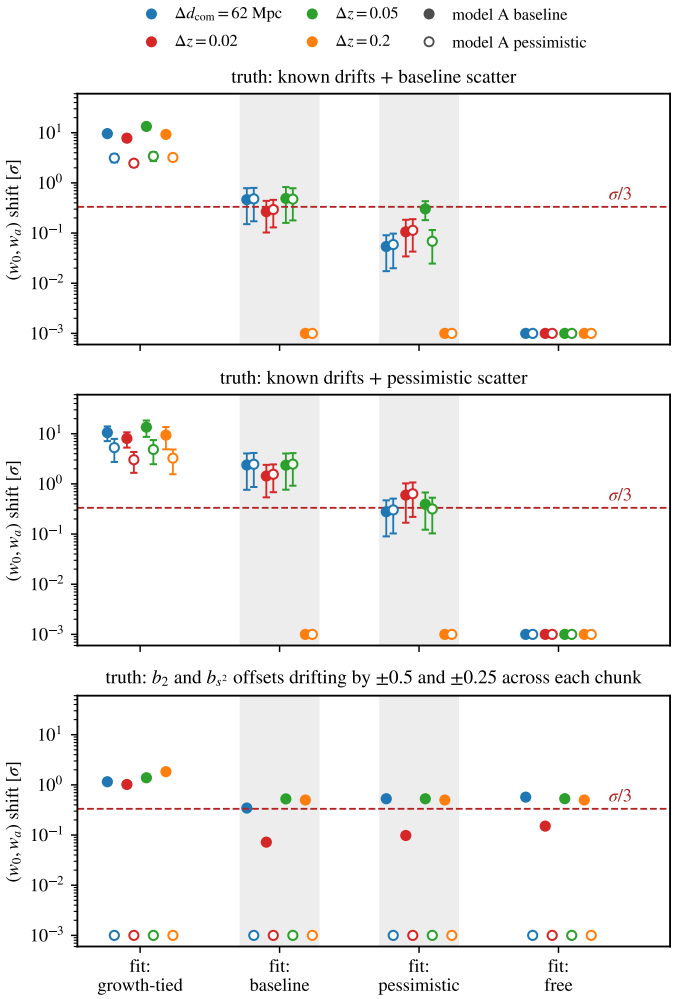


Figure 2: The accuracy test under model A: the (w_0, w_a) shift of each fit in units of its own marginalized ellipse, under every linear-bias prior (horizontal axis). Filled: model A baseline; open: model A pessimistic; colors: the layouts. Top and middle: 2000 truths with the known drifts plus residual scatter from the baseline prior (top) or the pessimistic prior (middle); points: the mean shift; bars: the 16th–84th percentiles. Bottom: the calibration-drift truth. Dashed line: $\sigma/3$. Gray columns: the baseline and pessimistic priors. Points at 10^{-3}: exact zeros. $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

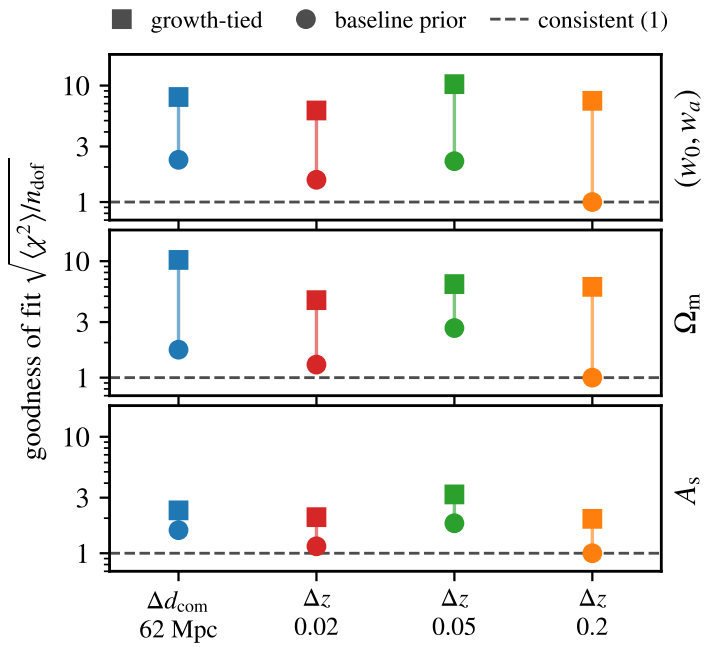


Figure 3: The goodness of fit of Eq. (5) under model A baseline, for truths with the known drifts plus pessimistic scatter (2000 draws): (w_0, w_a), Ω_m and A_s, per layout. Squares: the growth-tied prior; circles: the baseline prior. Dashed line: 1. Omitted: the free prior, exactly 1; the pessimistic prior, at most 1.023 in (w_0, w_a), 1.042 in Ω_m and 1.015 in A_s; on $\Delta z = 0.2$ the baseline prior is 1. $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

A referee who rejects either assumption of the reasonable configuration moves to the pessimistic one, and nothing needs retracting. The growth-tied prior is excluded under both variants, and the free prior under model A pessimistic (about 8, 2.5, 0.6 and 0.15) is the lower limit.

Limits. The covariance carries linear bias only, and its bias needs the update listed in the TODO below; comparisons between priors and variants hold it fixed. Shifts and goodness of fit are linear responses; the truths are theory, not mocks, and only a linear calibration drift is tested. Two counterterm questions are open (Appendix D): its template draws more signal from beyond the cut than the cut allows the linear model, and at negative α_0 the galaxy power turns negative within reach of a chain. Model B ($k_{\rm max} \approx 0.18\,h/{\rm Mpc}$ at $z = 0$, redshift-space loops) is not done. Appendix F lists every limitation.

TODO

- **Update the DESI covariance.** Its bias must be the data vector's $b_i = B_g/D(\bar{z}_i)$; the covariance adapter (`covariance/lssty1xdesi_covariance.py`, function `_lens_bias`) currently passes the raw chunk amplitude B_g, which understates the high-redshift variances. Then regenerate the DESI covariances and rerun the FoMs: the absolute

FoMs will drop by about 2.4–3.2 times.

- **Quote LSST Y1 3x2pt in Figure 1(b).** Compute the LSST Y1 3x2pt FoM$(\Omega_m, 10^9 A_s)$ from its Fisher matrix, with the same parameters and units as panel (b). Draw it in panel (b) as a horizontal dashed line, switch panel (a)'s LSST reference to the same dashed style so both panels match, and quote both values in the caption.

A Implementation and the Fisher construction

A.1 Model A in the likelihood

Model A runs in the project likelihood (`likelihood/desi_cl.py`) when its option `higher_order_bias` is set to 1. The example yamls `EXAMPLE_EVALUATE_MODEL_A1–A4.yaml` set it up on the four layouts, in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2.

Every model A parameter is per bin in the likelihood ($b_{1,i}$ and the offsets, $\alpha_{0,i}$ and $\mathrm{SN}_{0,i}$ of bin i); the grouping, one value per group or a value plus a tilt, is a statement of the parameter yamls, generated by `scripts/make_desi_params.py --model-a`, exactly as for the linear bias.

- **Quadratic and tidal bias.** The likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ with Eq. (4) and passes them to CosmoLike through the setter that takes the tidal bias per bin. The third-order bias is zero.

- **Counterterm.** CosmoLike's k^2 slot is $2\,b_1 b_K\,k^2 P_{\rm mm}$, with b_K its per-bin derivative-bias coefficient and k in units of H_0/c. The likelihood sets $b_K = \alpha_0/[2\,b_1 (c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}$Mpc. The counterterm therefore multiplies the nonlinear matter spectrum, CosmoLike's convention, where DESI's definition multiplies the linear one (subsection "The draft's data vector against CosmoLike" of `notes/higher_order_bias`).

- **Stochastic term.** A constant three-dimensional power projects exactly, without the Limber approximation, to an ℓ-independent spectrum,

$$C_\ell^{ij}\big|_{\rm stoch} = \mathrm{SN}_0\, P_{ij}\, K_{ij}, \qquad K_{ij} = \int dz\, n_i(z)\, n_j(z)\, \frac{dz/d\chi}{\chi^2}, \tag{6}$$

  with n_i the normalized Fingers-of-God-convolved redshift distribution of bin i and χ the comoving distance. The unit $P_{ii} = 1/\bar{n}_i$ is fixed by the bin's surface density, so that the projection with the raw distribution gives the angular shot noise. A cross pair of two bins of one tracer sample takes the mean of their two powers; a pair of two samples takes none, because different galaxies share no stochastic term.

- **Projection.** The low-rank non-Limber engine projects each bin's full one-loop $P_{gg,i}$, counterterm included, and the adjacent cross pairs carry the bilinear one-loop pair spectrum (subsection "Gaps against model A" of `notes/higher_order_bias`). Appendix D shows why the counterterm needs this engine.

A.2 The derivatives

The Fisher matrix needs the derivative of the data vector m (the band powers) with respect to every parameter. The notebook `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb` builds them in three groups.

Stencil derivatives. The seven cosmological parameters ($10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0, w_a), the six group amplitudes B_g and the velocity dispersions come from a five-point stencil of the model A likelihood, as in v1's notebook.

Per-bin operator columns. The four columns $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$ and $\partial m/\partial \alpha_{0,i}$ of every bin, each at fixed values of the other three, come from central differences through

the likelihood's bias setter. A wrapper intercepts the per-bin biases the likelihood passes to CosmoLike and shifts one of them for a set of bins. The steps are 1% of b_1, 0.02 in b_2 and b_{s^2}, and $5\,h^{-2}\,\mathrm{Mpc}^2$ in α_0. The set of bins is one color of a greedy coloring of the cross-pair graph: two bins joined by a measured cross pair never share a color. Each spectrum then involves at most one shifted bin, so one pair of evaluations gives the columns of every bin of that color. Every layout needs four colors, so each operator costs eight evaluations instead of two per bin.

Stochastic columns. The SN_0 columns are the exact templates $P_{ij}K_{ij}$ of Eq. (6); the vector is linear in SN_0.

Model A ties b_2 and b_{s^2} to b_1, so the derivative with respect to a bin's linear bias is the chain-rule sum

$$\left.\frac{dm}{db_{1,i}}\right|_{\mathrm{A}} = \frac{\partial m}{\partial b_{1,i}} + b_2^{\mathrm{L}\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}, \tag{7}$$

with $b_2^{\mathrm{L}\prime}$ the slope of the Lazeyras fit. Summed over a group with weights $1/D(z_i)$, these columns must reproduce the stencil derivative of the amplitude B_g (Appendix C).

A.3 The variants and the Fisher matrices

The parameters are, in order: the cosmological parameters and the velocity dispersions (with v1's Gaussian priors), the variant's higher-order parameters, and one amplitude $\beta_i = b_{1,i}\,D(z_i)$ per bin, fiducial $\beta_i = B_g$, as in v1. The column of β_i is Eq. (7) divided by $D(z_i)$. The higher-order columns of group g are sums over its bins:

$$\frac{\partial m}{\partial \Delta_{2,g}} = \sum_{i\in g}\frac{\partial m}{\partial b_{2,i}}, \qquad \frac{\partial m}{\partial t_{2,g}} = \sum_{i\in g}(\bar{z}_i - \bar{z}_g)\,\frac{\partial m}{\partial b_{2,i}}, \tag{8}$$

and likewise for Δ_s and α_0, with $t_{2,g}$ the tilt of Δ_2 in group g; the SN_0 column sums the group's stochastic templates. The exact-calibration configuration of Appendix E has no higher-order column; model A baseline has the offset columns with their Gaussian priors; model A pessimistic adds the tilt columns, with flat priors, for every group of more than one bin.

The data Fisher matrix is $F_{\mathrm{data}} = \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D}$ on the kept entries, with D the matrix of the columns above and C the Gaussian covariance computed for 14,000 square degrees, its inverse rescaled to the survey's 12,300. The Gaussian priors add to its diagonal. The higher-order parameters stand in front of the per-bin amplitudes and are marginalized like the cosmology, so v1's ladder code acts unchanged on the amplitude block. Every Fisher matrix is inverted through a pseudo-inverse (Appendix C).

A.4 The ladder, the response and the truths

- **Linear-bias priors.** Growth-tied sums the amplitude columns of each group into one; the baseline and pessimistic priors add v1's projected precision of the residual prior to the amplitude block (v1's appendices "How the prior enters the Fisher matrix" and "The memory projection, step by step"); free adds nothing.

- **Response.** A truth that moves the data by δm moves the best fit by $\delta p = F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\delta m$, to first order; the (w_0, w_a) shift is $\Delta_w = (\delta w^{\mathsf{T}}\mathsf{C}_w^{-1}\delta w)^{1/2}$, with C_w the fit's marginalized (w_0, w_a) covariance, and the Ω_m and A_s shifts are $|\delta p|/\sigma_p$ (v1's appendix "The shift formula").

- **Known drifts.** v1's two drift laws (appendix "The drift laws"): a conserved population in every group and the $r < 19.5$ flux limit in BGS. Under model A a drift δ_i of the linear bias moves b_2 and b_{s^2} as well: $\delta m = \sum_i (\partial m/\partial \beta_i)_{\rm A}\, \beta_i\, \delta_i$, with the model A column of Eq. (7).
- **Random truths and goodness of fit.** v1's 2000 draws per scatter family, with v1's seed and order; the goodness of fit of Eq. (5), $\sqrt{\langle \chi^2_w \rangle/2}$ for (w_0, w_a) and $\sqrt{\langle \chi^2_p \rangle}$ for one parameter p (v1's appendices "The goodness of fit" and "The same test for Ω_m and A_s").
- **Calibration drift.** One deterministic truth that moves only the offsets: $\delta m = \sum_i [(\partial m/\partial b_{2,i})\, \Delta_{2,i} + (\partial m/\partial b_{s^2,i})\, \Delta_{s,i}]$, with $\Delta_{2,i} = (\bar{z}_i - \bar{z}_g)/(\bar{z}_{\rm max} - \bar{z}_{\rm min})$ and $\Delta_{s,i} = \Delta_{2,i}/2$, where $\bar{z}_{\rm max}$ and $\bar{z}_{\rm min}$ are the largest and smallest mean redshifts of the group's bins. The drift therefore changes by 1 in Δ_2 and 0.5 in Δ_s from the group's first bin to its last, has zero mean over the group's bins, and vanishes in one-bin groups.

Every figure lives in its own folder under `plots/`, named after the figure, with its data (`data.json`, written by the notebook), one Python script that draws it, and the PDF the note includes. Each figure redraws from its folder alone.

B The known drifts, the goodness of fit and the calibration drift

Three figures carry the results of this appendix: Figure 4 (the known drifts), Figure 3 in the Summary (the goodness of fit) and Figure 5 (the calibration drift). The full numbers, every prior, both variants and both truths, are in each figure's `data.json` (`plots/drift_shifts`, `plots/goodness_of_fit`, `plots/calibration_drift`) and in the printed tables of `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb`. All use $k_{\rm max} = 0.075\, h/{\rm Mpc}$ quoted at $z = 0$, with each bin cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask; values below 10^{-9} are numerical zeros.

B.1 The known drifts

Figure 4 draws the shift of each known drift alone. The free prior absorbs both drifts exactly. Two readings. The growth-tied prior stays excluded: the conserved-population drift moves its (w_0, w_a) by 7.9–11σ under model A baseline and 2.5–3.4σ under model A pessimistic. The baseline and pessimistic priors absorb both drifts below $\sigma/3$ in every score; the largest entry, 0.32σ, is the BGS flux limit on $\Delta z = 0.05$ (pessimistic prior, model A baseline), a layout that resolves BGS into only six bins.

B.2 The goodness of fit

Figure 3, in the Summary, draws the goodness of fit of Eq. (5) under model A baseline and pessimistic scatter; its `data.json` holds every prior, both variants and both truth families. A value of 1 means the error covers the truth as often as it claims; below 1 the error is conservative. The free prior absorbs every random truth exactly.

Two readings beyond the Summary. The growth-tied fit's A_s error is the least damaged of its scores (a goodness of fit of 1.1 on $\Delta d_{\rm com} = 62$ Mpc under baseline scatter), but its (w_0, w_a)

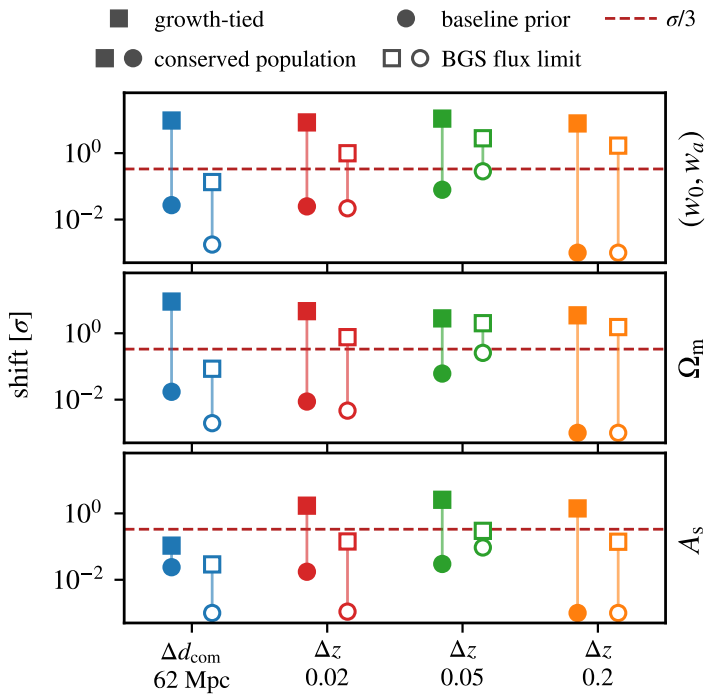


Figure 4: The shift of a model A baseline fit under each known drift: (w_0, w_a) in units of the fit's ellipse, Ω_m and A_s in their own marginalized σ, per layout. Squares: the growth-tied prior; circles: the baseline prior. Filled: the conserved-population drift; open: the BGS $r < 19.5$ flux limit. Dashed red line: $\sigma/3$. Points at 10^{-3}: values below it. Omitted: the pessimistic prior, at most 0.318σ in (w_0, w_a), 0.304 in Ω_m and 0.182 in A_s; the free prior, about 10^{-11}; model A pessimistic, whose growth-tied fit moves by 2.5–3.4σ in (w_0, w_a) under the conserved drift and whose baseline and pessimistic priors stay below 0.103σ in every score. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

error, at least 2.0 times too small, excludes it on every layout. And under model A pessimistic the growth-tied fit's (w_0, w_a) goodness of fit falls to 2.0–4.4, against 5.6–10.3 under model A baseline. A likely reason: the flat-prior tilts of the higher-order offsets can follow part of a linear-bias drift. That narrows the gap without making the growth-tied fit honest.

B.3 The calibration drift

Figure 5 draws the shifts of the calibration-drift truth under model A baseline. Model A pessimistic absorbs the drift under every prior and on every layout (every shift at most 1.6×10^{-12}) and is omitted. Under model A baseline the baseline, pessimistic and free priors exceed $\sigma/3$ in (w_0, w_a) on three layouts, and in A_s on $\Delta z = 0.05$ (0.37–0.41σ).

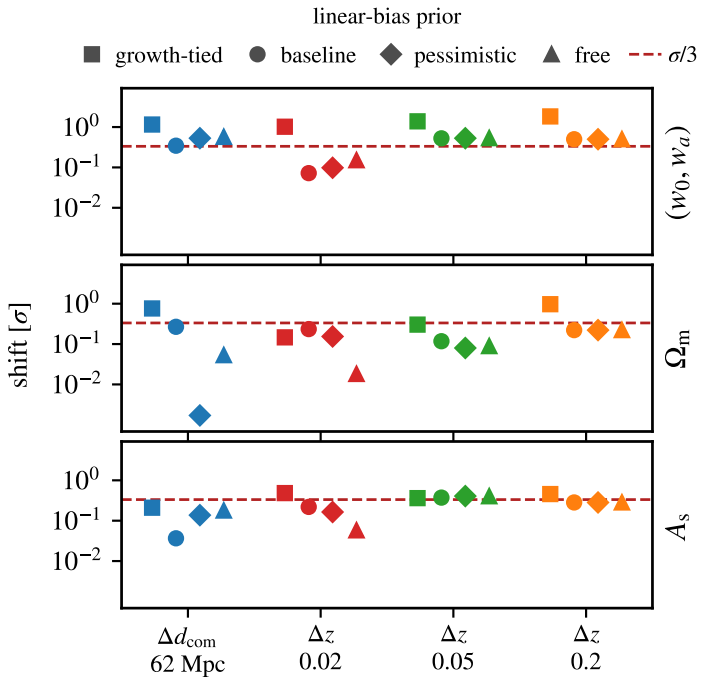


Figure 5: The shift of a model A baseline fit under the calibration drift (Δ_2 from about -0.5 to $+0.5$ and Δ_s from about -0.25 to $+0.25$ across each group): (w_0, w_a), Ω_m and A_s, per layout. Markers: the four linear-bias priors (square growth-tied, circle baseline, diamond pessimistic, triangle free). Dashed red line: $\sigma/3$. Omitted: model A pessimistic, which absorbs the drift exactly (every shift at most 1.6×10^{-12}). $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

C Numerical checks

- **The chain rule.** For every group, the per-bin columns of Eq. (7), summed with the weights $1/D(z_i)$, reproduce the stencil derivative of the group amplitude B_g to 9.8×10^{-8}, 1.2×10^{-7}, 4.0×10^{-8} and 1.3×10^{-8} of the stencil column's largest entry (layout order). The two independent routes, operator columns through the bias setter and the full likelihood through the stencil, agree.

- **The matched-prior identity.** When a fit's prior equals the truth's scatter, the projection identity $\mathsf{P}\Sigma\mathsf{P} = \mathsf{P}$ of v1's appendix "The goodness of fit" makes the scatter and noise terms of $\langle\chi^2_w\rangle$ sum to exactly 2. Over every layout and variant, $\max|\langle\chi^2_w\rangle - \Delta^2_{\rm drift} - 2| = 4.0 \times 10^{-10}$.

- **Exact recovery.** The free prior must absorb any truth of the linear bias. Its largest mean (w_0, w_a) shift over the random truths is 2.6×10^{-11}.

- **One direction without information.** On $\Delta d_{\rm com} = 62$ Mpc at this cut only 7 of the 18 BGS bins keep any band. Under model A pessimistic on the free prior, the flat-prior BGS tilts of Δ_2, Δ_s and α_0 then leave one combination of parameters with no information. A plain matrix inverse amplified round-off along it into a spurious 2σ shift. The notebook therefore inverts every Fisher matrix through a pseudo-inverse: it scales the matrix to unit diagonal, diagonalizes it, and drops the directions whose eigenvalue is below 10^{-12} of the largest. Exactly one of the 48 systems (four layouts, four linear-bias priors, and three higher-order configurations: model A baseline, model A pessimistic and the exact calibrations of Appendix E) drops one direction, and that direction's largest component on (w_0, w_a), Ω_m and A_s is 5×10^{-12}: a pure nuisance combination. The FoMs are unchanged.

- **The α_0 step.** At the negative stencil point, $\alpha_0 = -5\,h^{-2}\,{\rm Mpc}^2$, the factor $b_1^2 + \alpha_0 k^2$ of the linear and counterterm pieces turns negative above $k \approx 0.6\,h/{\rm Mpc}$, where the low-rank engine floors $\sqrt{P_{gg}}$. On $\Delta z = 0.05$, a step of 5 against a step of 1 changes the α_0 columns by at most 3.5×10^{-4} of each column's largest entry. Times a one-sigma α_0 ($88.9\,h^{-2}\,{\rm Mpc}^2$), the difference is worth $\Delta\chi^2 = 2.7 \times 10^{-4}$ in the worst bin and 1.0×10^{-3} with every bin shifted at once.

- **Reproducibility.** A recomputation of the $\Delta z = 0.05$ derivatives is bitwise identical to the cache.

- **The loop templates.** With $b_3 = 0$, CosmoLike's C-FAST-PT loop templates and Python FAST-PT agree to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ on the $\Delta d_{\rm com} = 62$ Mpc vector, pairs included (subsection "Gaps against model A" of `notes/higher_order_bias`).

- **The same data as v1.** The kept band powers are read from v1's cut cache at $k_{\rm max} = 0.075\,h/{\rm Mpc}$, and the v1 FoMs of Appendix E from v1's figure data.

D The counterterm: engines, scale cut and sign

Three properties of the counterterm $\alpha_0 k^2 P_{\rm mm}$ were measured on the $\Delta z = 0.05$ layout at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ ($z = 0$ value). The first is settled and fixes how the forecast is computed;

the other two are open.

D.1 The response needs the low-rank engine

CosmoLike has two non-Limber engines for C_ℓ^{gg}. FKEM, the FFTLog method of Fang et al. [8], projects the linear term exactly and moves everything nonlinear, the counterterm included, through its Limber difference, so its one-loop content is Limber. The low-rank engine factorizes each bin's full one-loop P_{gg}, less the white-noise constant of the loops (which Limber projects exactly), and projects it without the Limber approximation.

For the counterterm the difference is large. The low-rank α_0 response, $\partial C_\ell/\partial\alpha_0$, is 7.3 times the Limber one at $\ell = 20$, 1.8 times at $\ell = 50$, 1.26 times at $\ell = 100$ and about 1.05 times at $\ell = 200$–400 (median over the kept auto spectra). The reason is the radial modes of thin bins. Limber evaluates the power at the transverse wavenumber $k = (\ell + 1/2)/\chi$ only. A thin bin with sharp edges also passes radial modes of higher k, and the k^2 weight of the counterterm amplifies them; at low ℓ, where the transverse wavenumber is small, they dominate the response.

The low-rank response is converged: raising `nonlimber_accuracyboost` from 32 to 128 changes it by $\Delta\chi^2 = 6 \times 10^{-6}$, the rank from 5 to 10 by 5×10^{-14}, and the overall accuracy boost from 1 to 2 by 8×10^{-4}. For a one-sigma α_0 the two engines' responses differ by $\Delta\chi^2 \approx 90$, while at $\alpha_0 = 0$ they agree on the full vector to $\Delta\chi^2 = 3.6$. The forecast uses the low-rank engine.

D.2 Open: the counterterm's share beyond the cut

Apply the ultraviolet fraction of Eq. (1) to the α_0 template $\partial C_\ell/\partial\alpha_0$ instead of the band power. On the kept bands it is a median 0.076 and a maximum 0.14, and every kept band is above 0.05. For the fiducial model A vector the same statistic is a median 0.017 and a maximum 0.052.

The cut was built on the linear model, which keeps every band below 5%. The counterterm template draws more of its signal from $k > k_{\rm max}/D(\bar{z}_i)$, the scales the cut is meant to exclude: part of what the forecast learns about α_0 comes from there. Whether to rebuild the cut on the one-loop model, its templates included, is an open decision for the PI.

D.3 Open, for chains: negative power at negative α_0

The linear and counterterm pieces of Eq. (3) combine to $(b_1^2 + \alpha_0 k^2)\, P_{\rm mm}$, which is negative for $\alpha_0 < 0$ above $k \approx b_1/\sqrt{|\alpha_0|}$. At $\alpha_0 = -88.9\, h^{-2}\,{\rm Mpc}^2$, one prior width below zero, $P_{gg} - P_\infty$ (the spectrum the low-rank engine factorizes, with P_∞ the white-noise constant of the loops) turns negative above $k \approx 0.14\, h/{\rm Mpc}$, and the engine floors 47% of its grid. Its vector then curves in α_0: over $\pm 88.9\, h^{-2}\,{\rm Mpc}^2$ the departure from a straight line is worth $\chi^2 = 18.9$, while the FKEM vector is exactly linear (2×10^{-24}).

This is harmless for the Fisher forecast, which is evaluated at $\alpha_0 = 0$ with small steps (Appendix C). Chains that explore α_0 far below zero need one of the remedies the higher-order note gives for its white-noise constant (subsection "The white-noise limit of the renormalized loops"): a taper that returns P_{gg} to $b_1^2 P_{\rm mm}$ above the scale where one-loop perturbation theory is trusted, or a positivity prior that rejects points with $P_{gg} < 0$ below a chosen k.

E Comparison with linear bias and with exact calibrations

Neither comparison in this appendix is a result of this note. The linear-bias FoMs of v1 come from a model that `notes/linear_bias_kmax` shows is already biased at this cut, and the exact-calibration FoMs assume that b_2 and b_{s^2} are known exactly, which no forecast can assume. They appear here only to locate model A's FoMs. Both sets of FoMs are in `plots/baseline_ladder/data.json` (the v1 values and those of model A calibrated) and in the notebook's printed ladder table.

Linear bias. Against v1's linear bias, model A baseline keeps about 84, 86, 71 and 44% of v1's FoM(w_0, w_a) on the baseline prior, and model A pessimistic about 75, 73, 53 and 30% on the pessimistic prior (layouts in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2). The loss grows as the bins widen. A likely reason: on the wide layouts a group holds few bins, so one offset per group is close to one offset per bin, and the new parameters compete with each bin's linear bias for the same band powers. v1's FoMs remain the linear-bias reference.

Exact calibrations. With the calibrations taken as exact, the FoM exceeds v1's under every prior and on every layout: on the baseline prior it is about 1.3 times v1's on $\Delta d_{\rm com} = 62$ Mpc and about 1.9 times on $\Delta z = 0.05$. The loop terms are then not free: with $b_{2,i}$ and $b_{s^2,i}$ fixed by the calibrations at $b_{1,i}$, and $\alpha_0 = {\rm SN}_0 = 0$, the one-loop correction is a known function of $b_{1,i}$ and of the matter power spectrum. The loop terms grow as the fourth power of the growth factor and the linear term as b_1^2 times its square, so the size of the correction relative to the linear term carries information on the amplitude of the matter fluctuations that a linear-bias model does not have.

That information is only as good as the calibrations, and it is why this configuration is not a forecast. The offsets of model A baseline, with widths set by the known scatter of the calibrations, and its α_0 and ${\rm SN}_0$ remove it, and model A baseline sits below v1 under every prior and on every layout. The calibration drift of the Summary shows what trusting the calibrations costs: with exact calibrations it shifts (w_0, w_a) by 1.4–3.5σ under the growth-tied prior and by 0.19–1.01σ under the baseline, pessimistic and free priors. The notebook also prints this configuration's known-drift shifts and goodness of fit; they are not repeated here.

F Limitations

- **The covariance.** It is Gaussian and carries linear bias only. It also carries v1's mismatch: the covariance adapter (`_lens_bias`) uses each group's raw bias while the data vector uses $B_g/D(\bar{z}_i)$, which understates the high-redshift variances and inflates every absolute FoM by about 2.4–3.2. The comparisons between priors and variants hold the covariance fixed.

- **Linear response.** The shifts and the goodness of fit are first order in the truth's departure from the fiducial, as in v1. v1 estimates the dropped second-order term at a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift.

- **Theory truths, not mocks.** The project is a theory paper. The truths are the derived drifts, Gaussian residuals of the two stated sizes, and one linear drift of the

calibrations. Residuals rougher than the pessimistic prior, and calibrations that drift nonlinearly within a group, are untested.

- **The counterterm.** Its template's share beyond the cut and the negative power at negative α_0 are open (Appendix D).
- **Real-space loops.** Model A adds the real-space one-loop terms to linear Kaiser distortions. Model B, at $k_{\rm max} \approx 0.18\,h/\mathrm{Mpc}$ ($z = 0$) with the redshift-space one-loop terms, the radial counterterm and the scale-dependent stochastic term, is not done.
- **Third-order bias.** $b_3 = 0$, as in DESI's baseline; CosmoLike's b_3 template is a fixed table, acceptable only while $b_3 = 0$ (`notes/higher_order_bias`).

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